SIMATS ENGINEERING

ASSESSMENT TOOL 2: Research Review & Analysis

COURSE NAME: **ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS** **COURSE CODE:** **MLA01**

COURSE OUTCOME COVERED
CO4: *Develop intelligent software agents using suitable agent architectures and learning techniques.*(BTL 3)
Assessment Tool Weightage: 30%

Total Marks: 100 **Time: 90 Mins**
No. of Questions: 1

Annexure A

Research Review & Analysis

Code of Conduct:
I P Surendra Kumar Reddy (192425224) *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*
Signature of the student: P Surenda Kumar Reddy

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Domain Understanding	10%	
2. Literature Search and Source Selection	10%	
3. Literature Review and Synthesis	15%	
4. Comparative Analysis of AI Techniques	10%	
5. Critical Analysis of Research Findings	10%	
6. Research Gap Identification	10%	
7. Proposed Research Direction	10%	
8. Research Methodology / Analysis Framework	5%	
9. Experimental / Evidence-Based Validation	10%	
10. Result Analysis and Interpretation	5%	
11. Research Paper Quality and Referencing	10%	
12. Technical Presentation and Research Defense	5%	
Total	100%	

Signature of the Faculty Total Marks Awarded

ANSWER ANY ONE QUESTION

S.No.	Research Review & Analysis Question	Primary Topic
1	**Decision Tree Learning:** Review recent research on ID3, C4.5, and CART. Compare their attribute-selection methods, pruning strategies, interpretability, computational complexity, and classification performance. Identify a research gap and suggest a possible improvement.	Decision Trees
2	**Neural Networks vs Statistical Learning:** Review and compare research on neural-network and statistical-learning approaches for prediction and classification. Analyze their performance, data requirements, generalization, interpretability, computational cost, and suitability for different applications.	Statistical Learning & Neural Networks
3	**Kernel Methods:** Analyze research on SVM and kernel methods for non-linear classification. Compare linear, polynomial, and RBF kernels and evaluate their advantages and limitations compared with neural-network-based approaches.	Kernel Methods
4	**Reinforcement Learning:** Review research on Q-Learning, SARSA, and Deep Q-Networks. Compare their learning mechanisms, exploration strategies, convergence, computational requirements, and real-world applications. Identify challenges and research gaps.	Reinforcement Learning
5	**Explainable and Generalizable AI:** Review research on Inductive Learning and Explanation-Based Learning and analyze how these approaches contribute to generalization and explainability in AI. Identify current limitations and propose future research directions.	Inductive & Explanation-Based Learning

Structure of the Report:

S.No.	Paper Section	Expected Content
1	**Title**	Specific and concise title related to the selected research question/topic.
2	**Abstract**	Brief summary of the research problem, review methodology, major findings, research gap, and conclusion.
3	**Keywords**	4–6 important keywords related to the selected topic.
4	**1. Introduction**	Introduce the AI topic, background, importance, motivation, and scope of the review.
5	**2. Research Problem and Objectives**	Clearly state the research problem and specific objectives of the review.
6	**3. Literature Review**	Review relevant research papers and summarize major approaches, methodologies, findings, and contributions.
7	**4. Comparative Analysis**	Compare the reviewed approaches based on suitable parameters such as accuracy, complexity, interpretability, computational cost, scalability, and generalization.
8	**5. Critical Analysis**	Discuss strengths, weaknesses, limitations, assumptions, and practical challenges identified across the reviewed studies.
9	**6. Research Gap**	Identify unresolved problems, limitations in existing research, and areas requiring further investigation.
10	**7. Proposed Research Direction**	Suggest possible solutions, improvements, or future research directions based on the identified gap.
11	**8. Discussion**	Discuss the overall insights obtained from the review and their implications for AI research and applications.
12	**9. Conclusion**	Summarize the major findings, key insights, research gap, and future direction.
13	**References**	List all research papers, books, datasets, websites, and other sources using a consistent citation style.

A Research Review and Comparative Analysis of ID3, C4.5, and CART Decision Tree Algorithms

Abstract

Decision tree algorithms are widely used in machine learning because they provide a simple and interpretable way to perform classification and prediction. Among the early and influential decision tree algorithms are ID3, C4.5, and CART. Although these algorithms share the same basic idea of recursively dividing data into smaller groups, they differ in their attribute-selection criteria, tree structures, pruning methods, computational requirements, and practical performance.

This report reviews and compares ID3, C4.5, and CART. ID3 uses information gain to select attributes, while C4.5 extends ID3 by using gain ratio and adding capabilities such as handling continuous attributes and pruning. CART generally produces binary trees and commonly uses the Gini index for classification together with cost-complexity pruning. Research shows that there is no single decision tree algorithm that is optimal for every dataset. Performance can vary according to data type, class distribution, noise, number of attributes, and parameter settings. The report identifies the need for adaptive decision tree methods that can select suitable splitting and pruning strategies according to dataset characteristics. A possible improvement is therefore proposed using adaptive split selection combined with validation-based pruning.

Keywords: Decision Tree, ID3, C4.5, CART, Pruning, Classification

1. Introduction

Decision tree learning is a supervised machine learning approach used for classification and prediction. A decision tree represents decisions in a hierarchical structure consisting of a root node, internal decision nodes, branches and leaf nodes. Each internal node represents a test on an attribute, while each leaf represents a final prediction or class.

Decision trees are attractive because their predictions can be explained through a sequence of decisions. This makes them useful in areas where model interpretability is important. However, different tree-learning algorithms use different methods to select attributes and control tree growth.

Three important decision tree algorithms are **ID3, C4.5, and CART**. ID3 was introduced by J. Ross Quinlan and uses information gain to select attributes. Quinlan's original work also discussed limitations of the basic method when dealing with noisy or incomplete information.

C4.5 was developed as a successor to ID3. It introduced gain ratio, support for numerical attributes and additional mechanisms for handling practical data problems and pruning.

CART, developed by Breiman, Friedman, Olshen and Stone, uses a different tree-building framework and is particularly associated with binary trees, Gini-based classification and cost-complexity pruning.

This report compares these three algorithms to understand their strengths, limitations and appropriate applications.

2. Research Problem and Objectives

2.1 Research Problem

Although ID3, C4.5 and CART are all decision tree algorithms, they do not make decisions in exactly the same way. Their different attribute-selection measures and pruning strategies can produce different tree structures and classification results.

A major challenge is therefore determining which algorithm is more appropriate for a particular dataset. An algorithm that performs well on one dataset may not produce the same performance on another dataset.

Another issue is the trade-off between **accuracy and interpretability**. A very large tree may classify training data well but can become difficult to understand and may overfit the training data.

Therefore, a systematic comparison of ID3, C4.5 and CART is useful for understanding their differences and identifying opportunities to improve decision tree learning.

2.2 Objectives

The objectives of this report are:

1. To review the development and working principles of ID3, C4.5 and CART.
2. To compare their attribute-selection methods.
3. To examine their pruning strategies.
4. To compare their interpretability.
5. To discuss their computational complexity.
6. To compare their classification performance based on existing research.
7. To identify limitations and research gaps.
8. To propose a possible improvement to decision tree learning.

3. Literature Review

3.1 ID3

ID3, or **Iterative Dichotomiser 3**, was developed by J. Ross Quinlan. The algorithm uses information theory to select the attribute that provides the greatest information gain at each stage of tree construction.

Information gain is based on entropy. An attribute that produces a large reduction in uncertainty is preferred for splitting the data.

A simplified expression for entropy is:

$$\mathbf{Entropy(S) = -\Sigma\, p_i \log_2(p_i)}$$

Information gain can then be expressed as:

$$\mathbf{Gain(S,A) = Entropy(S) - \Sigma_v\, (|S_v| / |S|)\, Entropy(S_v)}$$

ID3 is relatively simple and produces understandable decision trees. However, the original approach has limitations. In particular, information gain can favour attributes with many possible values. Research has identified this multi-valued attribute bias as an important weakness.

ID3 also has more limited support for numerical attributes and missing data than later algorithms such as C4.5.

3.2 C4.5

C4.5 was developed by Ross Quinlan as an extension and successor to ID3. It retained the basic decision-tree approach while introducing several improvements.

One of its most important changes is the use of **gain ratio** instead of relying only on information gain.

Gain ratio considers the intrinsic information of a split:

Gain Ratio = Information Gain / Split Information

This reduces the tendency of information gain to favour attributes with many possible values.

C4.5 also supports numerical attributes by finding suitable thresholds and includes mechanisms for dealing with missing values and pruning. Its trees can also be converted into understandable if-then rules.

Therefore, C4.5 can be viewed as a more practical and flexible development of ID3.

3.3 CART

CART stands for **Classification and Regression Trees**. It was developed by Breiman, Friedman, Olshen and Stone. The method covers both classification and regression problems.

For classification, CART commonly uses the **Gini index** as a splitting criterion.

A simplified Gini impurity measure is:

$$\mathbf{Gini(S) = 1 - \Sigma p_i^2}$$

A lower Gini value represents a purer node.

Unlike ID3 and C4.5, CART generally constructs **binary trees**, meaning that each internal node is divided into two branches. Research comparing tree algorithms identifies binary structure and Gini-based splitting as key characteristics of CART.

CART also uses **cost-complexity pruning** to control tree size. The original CART work gives substantial attention to pruning, tree size and computational efficiency.

3.4 Research Evidence

Study	Focus	Main Finding	Relevance to This Review
Quinlan (1986)	ID3	Introduced information-gain-based decision-tree learning	Foundation for ID3
Quinlan (1993)	C4.5	Extended ID3 with gain ratio and practical improvements	Foundation for C4.5
Breiman et al. (1984)	CART	Developed classification/regression tree framework and pruning	Foundation for CART
Martin (1995)	Tree splitting	Examined statistical issues in splitting/stopping	Supports critical analysis
Shamrat et al. (2022)	ID3/C4.5/CART comparison	Compared performance of the three approaches	Supports comparative discussion

These studies collectively demonstrate the evolution of decision-tree learning from information-gain-based splitting toward approaches that address attribute-selection bias, numerical data handling, pruning, and statistical reliability. However, the reviewed findings also indicate that comparative performance remains dataset-dependent, meaning that no single decision-tree algorithm is universally optimal.

4. Comparative Analysis

4.1 Attribute-Selection Methods

The main difference between the three algorithms is how they choose the attribute used for splitting.

Feature	ID3	C4.5	CART
Main criterion	Information Gain	Gain Ratio	Gini Index
Tree structure	Multiway	Multiway	Binary
Numerical attributes	Limited	Supported	Supported
Missing values	Limited	Better support	Supported through implementation strategies
Main objective	Reduce entropy	Improve information gain selection	Reduce impurity

ID3 selects the attribute with the highest information gain. C4.5 improves this approach by using gain ratio. CART uses Gini impurity for classification.

4.2 Pruning Strategies

Pruning is important because unrestricted tree growth can create overly complex trees.

ID3

The basic ID3 algorithm is primarily associated with growing a tree based on information gain and has limited built-in pruning compared with later approaches. This can make the resulting tree more vulnerable to overfitting.

C4.5

C4.5 includes **error-based post-pruning**. The tree is first constructed and then unnecessary branches can be removed to improve generalization.

CART

CART uses **cost-complexity pruning**. The objective is to balance the classification error with tree complexity and select an appropriately sized subtree. The original CART methodology explicitly develops minimal cost-complexity pruning.

Therefore, CART provides a particularly systematic approach to controlling tree complexity.

4.3. Interpretability

Interpretability is one of the major advantages of decision trees.

ID3

ID3 produces a tree that can be followed from the root to a leaf. This makes individual predictions relatively easy to explain.

C4.5

C4.5 also provides understandable tree structures and can transform trees into if-then rules, which can improve the presentation of classification decisions.

CART

CART is also highly interpretable, particularly when the tree is not excessively deep. Binary splits can make individual decision paths straightforward to follow.

However, interpretability decreases when the tree becomes very large. Therefore, pruning is important not only for predictive performance but also for making the model easier to understand.

4.4. Computational Complexity

The computational cost of a decision tree depends on factors such as the number of training samples, number of features, feature types and number of possible split points.

ID3 is conceptually simple, but evaluating information gain across attributes can become expensive as the dataset grows.

C4.5 performs additional calculations for gain ratio, numerical attributes, missing values and pruning. Therefore, it can require more computation than the basic ID3 approach.

CART also performs split searches and pruning. Its binary structure can produce deeper trees in some situations, although binary splitting can also provide a systematic search space.

Research on decision-tree splitting and computational complexity shows that the choice of splitting criterion and tree-construction strategy affects computational behaviour.

The important point is that computational efficiency cannot be judged only from the algorithm name. Dataset size, feature type and implementation also influence execution time.

Criterion	**ID3**	**C4.5**	**CART**	**Critical Observation**
Split criterion	Information Gain	Gain Ratio	Gini Index	Each criterion has different biases
Split structure	Multiway	Multiway	Binary	CART may produce deeper trees
Numerical data	Limited	Supported	Supported	C4.5/CART are more flexible
Pruning	Limited	Error-based	Cost-complexity	Pruning improves generalization
Interpretability	High	High	High	Interpretability decreases as tree size grows
Main weakness	Multi-value bias	Higher complexity	Binary-split limitation	No single method is universally optimal

The comparison shows that the development from ID3 to C4.5 addressed several limitations of the original information-gain approach. CART follows a different design based on binary splitting and cost-complexity pruning. Therefore, algorithm selection should depend on the characteristics of the dataset rather than assuming that one algorithm will always provide the best performance.

4.5. Classification Performance

Classification performance varies according to the dataset and experimental conditions.

A comparative study specifically examining ID3, C4.5 and CART reports that the algorithms can differ in their working processes and classification accuracy.

There is therefore no universal rule stating that one algorithm will always achieve the highest accuracy.

Performance can be affected by:

- Dataset size
- Number of features
- Class imbalance

- Missing values
- Numerical and categorical attributes
- Noise
- Overfitting
- Parameter settings
- Pruning strategy

C4.5 may be advantageous when the dataset contains numerical and missing values because of its extended handling capabilities. CART may be attractive when binary splitting and systematic pruning are desired. ID3 remains useful for understanding the basic principles of decision-tree learning because of its simple information-gain approach.

4.6. Overall Comparison

Criteria	**ID3**	**C4.5**	**CART**
Attribute selection	Information Gain	Gain Ratio	Gini Index
Tree type	Multiway	Multiway	Binary
Numerical data	Limited	Yes	Yes
Missing values	Limited	Better support	Implementation dependent
Pruning	Limited/basic	Error-based pruning	Cost-complexity pruning
Interpretability	High	High	High
Algorithm complexity	Lower	Higher than ID3	Moderate
Classification use	Yes	Yes	Yes
Regression use	No	Mainly classification	Yes
Main strength	Simplicity	Flexibility	Robust tree/pruning framework
Main weakness	Multi-value bias	More computational work	Binary splits may produce deeper trees

The comparison shows that **ID3 prioritizes simplicity, C4.5 prioritizes flexibility, and CART provides a strong framework for binary splitting and pruning**.

5. Critical Analysis

The comparison demonstrates that the evolution from ID3 to C4.5 was largely motivated by practical limitations of the original approach.

ID3's information gain criterion is simple but can favour attributes with many distinct values. This is a known limitation of the method.

C4.5 addresses part of this problem through gain ratio. However, research has shown that gain ratio itself is not completely free from statistical bias. Martin's analysis found that gain ratio only partially compensates for the multi-valued bias and that alternative splitting and stopping criteria can also have advantages.

CART takes a different approach by using binary splits and cost-complexity pruning. Its original framework includes detailed treatment of pruning, splitting rules, computational efficiency and model interpretation.

Therefore, the comparison suggests that improving one component of a decision-tree algorithm does not automatically solve every problem. A splitting criterion may perform well for one type of data but not another. Similarly, an aggressive pruning strategy may improve generalization but can remove useful structure.

The best decision-tree method should therefore consider the characteristics of the dataset rather than relying on one fixed criterion.

5.1 Strengths and Weaknesses of Existing Approaches

ID3 has the advantage of simplicity and provides an easy-to-understand foundation for decision-tree learning. However, its reliance on information gain can favour attributes with many possible values. C4.5 addresses this issue through gain ratio and provides additional capabilities such as numerical attribute handling and pruning. However, these additional capabilities increase the complexity of the algorithm.

CART provides a systematic binary-splitting framework and cost-complexity pruning. Its binary structure can simplify the splitting process but may result in deeper trees than multiway approaches. Therefore, each algorithm represents a different trade-off between simplicity, flexibility, computational requirements and model complexity.

6. Research Gap

Existing decision-tree algorithms generally use predefined splitting and pruning strategies. **ID3 uses information gain, C4.5 uses gain ratio, and CART commonly uses the Gini index.** Although these approaches have been extensively studied, their effectiveness can vary depending on dataset characteristics such as **class imbalance, feature cardinality, noise, and the proportion of numerical and categorical attributes**.

A further research opportunity is therefore to develop an **adaptive decision-tree framework** that does not rely on a single fixed splitting criterion. Instead, the framework could evaluate **dataset characteristics and validation performance** before selecting an appropriate splitting and pruning strategy.

Such an approach could address the trade-off between **predictive accuracy, computational cost, tree complexity, and interpretability**.

The proposed framework would evaluate candidate models using accuracy, precision, recall, F1-score, tree depth, number of nodes and computational time. A multi-objective evaluation could then be used to select a model that provides a suitable balance between predictive performance and interpretability.

7. Proposed Research Direction

A possible solution to the identified research gap is an Adaptive Decision Tree Selection and Pruning Framework. The framework would analyse dataset characteristics, including feature types, class imbalance, dataset size and feature cardinality. It would then evaluate candidate splitting criteria such as Information Gain, Gain Ratio and Gini Index.

Candidate decision trees would be pruned using suitable validation-based strategies and evaluated using accuracy, precision, recall, F1-score, tree depth, number of nodes and computational time. A multi-objective evaluation could then select the model that provides an appropriate balance between predictive performance, computational efficiency and interpretability.

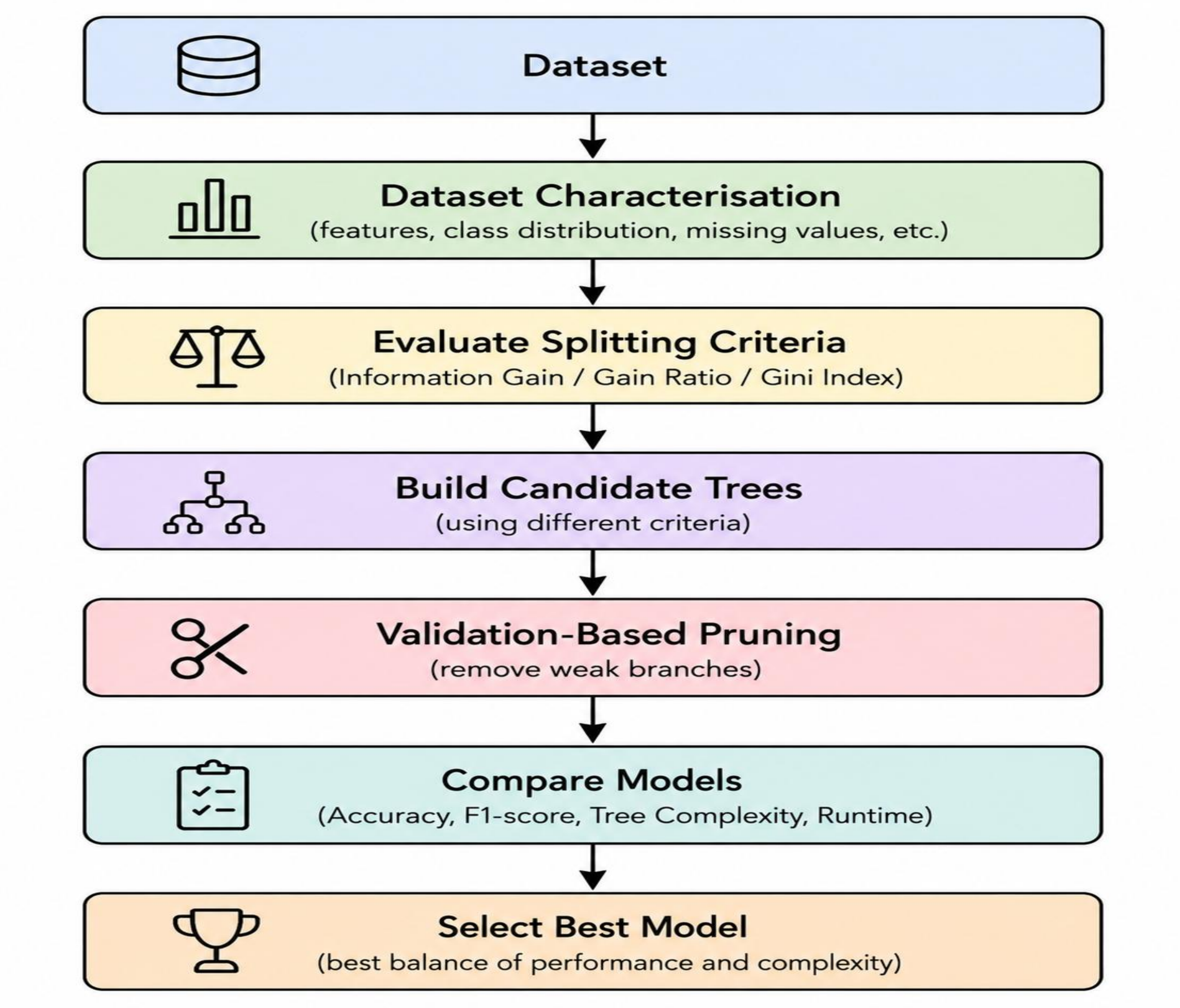

Figure 1: Proposed Adaptive Decision Tree Framework.

Proposed process

Step 1: Analyse the dataset.

Step 2: Identify feature types, class distribution and number of unique values.

Step 3: Evaluate candidate splitting criteria such as:

- Information Gain
- Gain Ratio
- Gini Index

Step 4: Build candidate trees using the selected criteria.

Step 5: Apply validation-based pruning.

Step 6: Compare the candidate models using:

- Accuracy
- Precision
- Recall
- F1-score
- Tree depth
- Number of nodes
- Computational time

Step 7: Select the model that provides the best balance between predictive performance and complexity.

This approach could reduce dependence on one fixed splitting criterion.

8. Discussion

The review shows that ID3, C4.5 and CART are closely related but make different design choices.

ID3 provides a useful foundation because of its simple information-gain approach. However, its limitations motivated improvements in later algorithms.

C4.5 extends ID3 by introducing gain ratio and improved handling of practical data problems. Its ability to work with numerical attributes and perform pruning makes it more flexible than the original ID3.

CART uses binary splitting and provides a systematic cost-complexity pruning framework. This makes it particularly useful for practical classification and regression tasks.

The comparison also shows that classification accuracy should not be considered the only measure of a decision tree. A model should also be evaluated according to interpretability, computational cost and tree complexity.

For example, a slightly less accurate tree may be preferable if it is significantly smaller and easier to explain. Similarly, a highly accurate model may not be appropriate if it overfits the training data.

Therefore, future decision-tree research should focus on balancing **accuracy, generalization, computational efficiency and interpretability**.

9. Conclusion

ID3, C4.5 and CART are three important decision-tree algorithms that demonstrate the evolution of tree-based machine learning.

ID3 uses **information gain** and provides a simple and interpretable approach, but it has limitations such as bias toward attributes with many possible values. C4.5 improves on ID3 by using **gain ratio**, supporting numerical attributes and incorporating pruning. CART uses **Gini-based splitting**, generally constructs binary trees and applies **cost-complexity pruning**.

The comparison shows that there is no universally best algorithm. The appropriate choice depends on the dataset, feature types, class distribution, noise, computational requirements and desired level of interpretability.

The main research gap identified is the need for more adaptive decision-tree approaches that can select suitable splitting and pruning strategies according to the characteristics of the dataset.

A possible future direction is therefore to develop an adaptive framework that evaluates multiple splitting criteria and pruning strategies and selects the best combination using validation performance and model complexity.

Such an approach could improve the balance between **classification accuracy, generalization, computational efficiency and interpretability**.

References

1. Quinlan, J. R. (1986). **Induction of Decision Trees**. *Machine Learning, 1*, 81–106. https://storm.cis.fordham.edu/~gweiss/selected-papers/quinlan86.pdf
2. Quinlan, J. R. (1993). **C4.5: Programs for Machine Learning**. Morgan Kaufmann. https://www.sciencedirect.com/book/monograph/9780080500584/c4-5
3. Breiman, L., Friedman, J. H., Olshen, R. A., & Stone, C. J. (1984). **Classification and Regression Trees**. Chapman & Hall. https://books.google.com/books/about/Classification_and_Regression_Trees.html?id=8k1DvQEACAAJ
4. Martin, J. K. (1995). **An Exact Probability Metric for Decision Tree Splitting and Stopping**. *Proceedings of the Fifth International Workshop on Artificial Intelligence and Statistics*. https://proceedings.mlr.press/r0/martin95b.html
5. Mehedi Shamrat, F. M. J., Ranjan, R., Hasib, K. M., Yadav, A., et al. (2022). **Performance Evaluation Among ID3, C4.5, and CART Decision Tree Algorithm**. *Pervasive Computing and Social Networking*, 127–142. https://doi.org/10.1007/978-981-16-5640-8_11
6. Liu, X. (2022). **Application of Decision Tree-Based Classification Algorithm on Content Marketing**. *Journal of Mathematics*. https://onlinelibrary.wiley.com/doi/10.1155/2022/6469054
7. Scikit-learn documentation. **Decision Trees — Tree Algorithms: ID3, C4.5, C5.0 and CART**. https://github.com/scikit-learn/scikit-learn/blob/main/doc/modules/tree.rst

Evaluation Rubrics:

Criterion	Weightage (%)	Excellent	Good	Average	Poor
1. Research Problem Identification and Understanding	10%	9–10% – Clearly identifies the research problem, objectives, scope, and relevance to AI learning approaches.	7–8% – Research problem and objectives are clearly identified with minor omissions.	5–6% – Basic understanding of the research problem with some gaps.	0–4% – Problem is unclear, inappropriate, or poorly understood.
2. Literature Search and Source Selection	10%	9–10% – Uses diverse, highly relevant, recent, and credible research papers, journals, conferences, and technical sources.	7–8% – Uses relevant and credible sources with minor limitations.	5–6% – Uses a limited number of relevant sources.	0–4% – Sources are insufficient, unreliable, or largely irrelevant.
3. Literature Review and Synthesis	15%	13–15% – Provides a comprehensive review and synthesizes findings across multiple studies rather than merely summarizing them.	10–12% – Good review with meaningful comparison of major studies.	7–9% – Basic review with limited synthesis and comparison.	0–6% – Superficial review with little understanding of existing research.

4. Analysis of AI Techniques / Methods	15%	13–15% – Provides technically accurate and in-depth analysis of relevant methods such as decision trees, statistical learning, neural networks, kernel methods, or reinforcement learning.	10–12% – Provides good technical analysis with minor gaps.	7–9% – Provides basic technical description with limited analysis.	0–6% – Methods are poorly understood or inaccurately described.
5. Comparative Analysis of Research Approaches	10%	9–10% – Critically compares approaches based on accuracy, complexity, interpretability, scalability, computational cost, and application suitability.	7–8% – Provides meaningful comparison using several parameters.	5–6% – Basic comparison with limited criteria.	0–4% – Little or no meaningful comparison.
6. Critical Evaluation of Research Findings	10%	9–10% – Critically evaluates strengths, weaknesses, assumptions, experimental methods, and findings of existing studies.	7–8% – Provides good critical evaluation with minor gaps.	5–6% – Some critical observations are provided but analysis is limited.	0–4% – Mostly descriptive with little critical evaluation.

7. Identification of Research Gaps	10%	9–10% – Clearly identifies significant research gaps, limitations, unresolved issues, and opportunities for further investigation.	7–8% – Identifies relevant research gaps with reasonable justification.	5–6% – Identifies basic gaps but provides limited justification.	0–4% – Research gaps are missing or incorrectly identified.
8. Research Insights and Recommendations	5%	5% – Provides insightful conclusions and technically feasible recommendations for future research or applications.	4% – Provides relevant conclusions and recommendations.	2–3% – Basic conclusions with limited recommendations.	0–1% – Conclusions or recommendations are absent or inappropriate.
9. Technical Report and Referencing	10%	9–10% – Report is well structured, technically accurate, professionally written, and uses a consistent citation/reference style.	7–8% – Well-structured report with minor writing or referencing issues.	5–6% – Adequate report but contains several structural or referencing issues.	0–4% – Poorly structured, inadequately referenced, or incomplete report.
10. Originality, Academic Integrity and Use of Sources	5%	5% – Demonstrates original synthesis, proper citation, and strong academic integrity with no inappropriate copying.	4% – Good originality and proper citation with minor issues.	2–3% – Some original analysis but significant dependence on sources.	0–1% – Evidence of substantial copying, poor citation, or academic-integrity concerns.